主理想形式层的代数化判据

Counting scope. The primary problem is Tag 0EJ9, marked below. All preceding numbered statements are supporting author context and are not counted as additional problems. This is a local pilot with a temporary ID, not production admission.

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Difficulty H2: 本题在有对偶化复形的 Noetherian 局部基底上，由 (2,3) 不等式证明主理想形式层可延拓至 X。证明以局部上同调的有限模块控制障碍，构造有限分次 A[T] 模块并由齐次生成元推出 Mittag–Leffler 性质。代数化收口判据和局部对偶估计可引用，但障碍有限性与稳定像论证仍由本条完成，判 H2。.

History: excerpt from revision local cached snapshot; exact chapter hashes; revision unknown, algebraization.tex, lines 6518–6654. Only reference presentation was adapted. Licensed under GNU FDL 1.2 or later; no invariant sections or cover texts. See accompanying COPYING.

Situation 0.1. Here A is a Noetherian ring and $I \subset \mathfrak{a} \subset A$ are ideals. We set $X = \mathrm{Spec}(A)$, $Y = V(I) = \mathrm{Spec}(A/I)$, and $Z = V(\mathfrak{a}) = \mathrm{Spec}(A/\mathfrak{a})$. Furthermore $U = X \setminus Z$.

Let Y be a Noetherian scheme and let $Z \subset Y$ be a closed subset. We define a function

$$\delta_Z^Y = \delta_Z : Y \longrightarrow \mathbf{Z}_{\geq 0} \cup \{\infty\} \tag{0.1.1}$$

which measures the "distance" of a point of Y from Z. For an informal discussion, please see Remark remark discussion (Tag 0EJ0). Let $y \in Y$. We set $\delta_Z(y) = \infty$ if y is contained in a connected component of Y which does not meet Z. If y is contained in a connected component of Y which meets Z, then we can find $k \geq 0$ and a system

$$V_0 \subset W_0 \supset V_1 \subset W_1 \supset \ldots \supset V_k \subset W_k$$

of integral closed subschemes of Y such that $V_0 \subset Z$ and $y \in W_k$ is the generic point. Set $c_i = \mathrm{codim}(V_i, W_i)$ for $i = 0, \ldots, k$ and $b_i = \mathrm{codim}(V_{i+1}, W_i)$ for $i = 0, \ldots, k-1$. For such a system we set

$$\delta(V_0, W_0, V_1, \ldots, W_k) = k + \max_{i=0,1,\ldots,k} (c_i + c_{i+1} + \ldots + c_k - b_i - b_{i+1} - \ldots - b_{k-1})$$

This is $\geq k$ as we can take $i = k$ and we have $c_k \geq 0$. Finally, we set

$$\delta_Z(y) = \min \delta(V_0, W_0, V_1, \ldots, W_k)$$

where the minimum is over all systems of integral closed subschemes of Y as above.

Definition 0.2. In Situation 0.1 let $(\mathcal{F}_n)$ be an object of $Coh(U, I\mathcal{O}_U)$. Let a, b be integers. Let δ_Z^Y be as in (0.1.1). We say $(\mathcal{F}_n)$ *satisfies the* (a,b)*-inequalities* if for $y \in U \cap Y$ and a prime $\mathfrak{p} \subset \mathcal{O}_{X,y}^\wedge$ with $\mathfrak{p} \notin V(I\mathcal{O}_{X,y}^\wedge)$

(1) if $V(\mathfrak{p}) \cap V(I\mathcal{O}_{X,y}^\wedge) \neq \{\mathfrak{m}_y^\wedge\}$, then

$$\mathrm{depth}((\mathcal{F}_y^\wedge)_\mathfrak{p}) + \delta_Z^Y(y) \geq a \quad \text{or} \quad \mathrm{depth}((\mathcal{F}_y^\wedge)_\mathfrak{p}) + \dim(\mathcal{O}_{X,y}^\wedge/\mathfrak{p}) + \delta_Z^Y(y) > b$$

(2) if $V(\mathfrak{p}) \cap V(I\mathcal{O}_{X,y}^\wedge) = \{\mathfrak{m}_y^\wedge\}$, then

$$\mathrm{depth}((\mathcal{F}_y^\wedge)_\mathfrak{p}) + \delta_Z^Y(y) > a$$

We say $(\mathcal{F}_n)$ *satisfies the strict* (a, b)*-inequalities* if for $y \in U \cap Y$ and a prime $\mathfrak{p} \subset \mathcal{O}^\wedge_{X,y}$ with $\mathfrak{p} \notin V(I\mathcal{O}^\wedge_{X,y})$ we have

$$\text{depth}((\mathcal{F}^\wedge_y)_\mathfrak{p}) + \delta^Y_Z(y) > a \quad \text{or} \quad \text{depth}((\mathcal{F}^\wedge_y)_\mathfrak{p}) + \dim(\mathcal{O}^\wedge_{X,y}/\mathfrak{p}) + \delta^Y_Z(y) > b$$

Lemma 0.3. *In Situation 0.1 let (M_n) be an inverse system of A-modules as in Lemma lemma system of modules (Tag 0DXT) and let $(\mathcal{F}_n)$ be the corresponding object of $Coh(U, I\mathcal{O}_U)$. Let $d \geq cd(A, I)$ and $s \geq 0$ be integers. With notation as above assume*

1. *A is local with maximal ideal $\mathfrak{m} = \mathfrak{a}$,*
2. *A has a dualizing complex, and*
3. *$(\mathcal{F}_n)$ satisfies the $(s, s + d)$-inequalities (Definition 0.2).*

Let E be an injective hull of the residue field of A. Then for $i \leq s$ there exists a finite A-module N annihilated by a power of I and for $n \gg 0$ compatible maps

$$H^i_\mathfrak{m}(M_n) \to \text{Hom}_A(N, E)$$

whose cokernels are finite length A-modules and whose kernels K_n form an inverse system such that $\text{Im}(K_{n''} \to K_{n'})$ has finite length for $n'' \gg n' \gg 0$.

证明. Let $\omega^\bullet_A$ be a normalized dualizing complex. Then $\delta^Y_Z = \delta$ is the dimension function associated with this dualizing complex. Observe that $\text{Ext}^{-i}_A(M_n, \omega^\bullet_A)$ is a finite A-module annihilated by I^n. Fix $0 \leq i \leq s$. Below we will find $n_1 > n_0 > 0$ such that if we set

$$N = \text{Im}(\text{Ext}^{-i}_A(M_{n_0}, \omega^\bullet_A) \to \text{Ext}^{-i}_A(M_{n_1}, \omega^\bullet_A))$$

then the kernels of the maps

$$N \to \text{Ext}^{-i}_A(M_n, \omega^\bullet_A), \quad n \geq n_1$$

are finite length A-modules and the cokernels Q_n form a system such that $\text{Im}(Q_{n'} \to Q_{n''})$ has finite length for $n'' \gg n' \gg n_1$. This is equivalent to the statement that the system $\{\text{Ext}^{-i}_A(M_n, \omega^\bullet_A)\}_{n\geq 1}$ is essentially constant in the quotient of the category of finite A-modules modulo the Serre subcategory of finite length A-modules. By the local duality theorem (Dualizing Complexes, Lemma lemma special case local duality (Tag 0AAK)) and Matlis duality (Dualizing Complexes, Proposition proposition matlis (Tag 08Z9)) we conclude that there are maps

$$H^i_\mathfrak{m}(M_n) \to \text{Hom}_A(N, E), \quad n \geq n_1$$

as in the statement of the lemma.

Pick $f \in \mathfrak{m}$. Let $B = A^\wedge_f$ be the I-adic completion of the localization A_f. Recall that $\omega^\bullet_{A_f} = \omega^\bullet_A \otimes_A A_f$ and $\omega^\bullet_B = \omega^\bullet_A \otimes_A B$ are dualizing complexes (Dualizing Complexes, Lemma lemma dualizing localize (Tag 0A7G) and lemma completion henselization dualizing (Tag 0DWD)). Let M be the finite B-module $\lim M_{n,f}$ (compare with discussion in Cohomology of Schemes, Lemma lemma inverse systems affine (Tag 087W)). Then

$$\text{Ext}^{-i}_A(M_n, \omega^\bullet_A)_f = \text{Ext}^{-i}_{A_f}(M_{n,f}, \omega^\bullet_{A_f}) = \text{Ext}^{-i}_B(M/I^nM, \omega^\bullet_B)$$

Since $\mathfrak{m}$ can be generated by finitely many $f \in \mathfrak{m}$ it suffices to show that for each f the system

$$\{\text{Ext}^{-i}_B(M/I^nM, \omega^\bullet_B)\}_{n\geq 1}$$

is essentially constant. Some details omitted.

Let $\mathfrak{q} \subset IB$ be a prime ideal. Then $\mathfrak{q}$ corresponds to a point $y \in U \cap Y$. Observe that $\delta(\mathfrak{q}) = \dim(\overline{\{y\}})$ is also the value of the dimension function associated to $\omega_B^\bullet$ (we omit the details; use that $\omega_B^\bullet$ is gotten from $\omega_A^\bullet$ by tensoring up with B). Assumption (3) guarantees via Lemma lemma elementary (Tag 0EJ4) that Lemma lemma algebraize local cohomology bis (Tag 0EFR) applies to $B_\mathfrak{q}, IB_\mathfrak{q}, \mathfrak{q}B_\mathfrak{q}, M_\mathfrak{q}$ with s replaced by $s - \delta(y)$. We obtain that

$$H^{i-\delta(\mathfrak{q})}_{\mathfrak{q}B_\mathfrak{q}}(M_\mathfrak{q}) = \lim H^{i-\delta(\mathfrak{q})}_{\mathfrak{q}B_\mathfrak{q}}((M/I^nM)_\mathfrak{q})$$

and this module is annihilated by a power of I. By Lemma lemma terrific (Tag 0EHB) we find that the inverse systems $H^{i-\delta(\mathfrak{q})}_{\mathfrak{q}B_\mathfrak{q}}((M/I^nM)_\mathfrak{q})$ are essentially constant with value $H^{i-\delta(\mathfrak{q})}_{\mathfrak{q}B_\mathfrak{q}}(M_\mathfrak{q})$. Since $(\omega_B^\bullet)_\mathfrak{q}[-\delta(\mathfrak{q})]$ is a normalized dualizing complex on $B_\mathfrak{q}$ the local duality theorem shows that the system

$$\mathrm{Ext}^{-i}_B(M/I^nM, \omega_B^\bullet)_\mathfrak{q}$$

is essentially constant with value $\mathrm{Ext}^{-i}_B(M, \omega_B^\bullet)_\mathfrak{q}$.

To finish the proof we globalize as in the proof of Lemma lemma bootstrap (Tag 0EFX); the argument here is easier because we know the value of our system already. Namely, consider the maps

$$\alpha_n : \mathrm{Ext}^{-i}_B(M/I^nM, \omega_B^\bullet) \longrightarrow \mathrm{Ext}^{-i}_B(M, \omega_B^\bullet)$$

for varying n. By the above, for every $\mathfrak{q}$ we can find an n such that α_n is surjective after localization at $\mathfrak{q}$. Since B is Noetherian and $\mathrm{Ext}^{-i}_B(M, \omega_B^\bullet)$ a finite module, we can find an n such that α_n is surjective. For any n such that α_n is surjective, given a prime $\mathfrak{q} \in V(IB)$ we can find an $n' > n$ such that $\mathrm{Ker}(\alpha_n)$ maps to zero in $\mathrm{Ext}^{-i}(M/I^{n'}M, \omega_B^\bullet)$ at least after localizing at $\mathfrak{q}$. Since $\mathrm{Ker}(\alpha_n)$ is a finite A-module and since supports of sections are quasi-compact, we can find an n' such that $\mathrm{Ker}(\alpha_n)$ maps to zero in $\mathrm{Ext}^{-i}(M/I^{n'}M, \omega_B^\bullet)$. In this way we see that $\mathrm{Ext}^{-i}(M/I^nM, \omega_B^\bullet)$ is essentially constant with value $\mathrm{Ext}^{-i}(M, \omega_B^\bullet)$. This finishes the proof. □

Lemma 0.4. *In Situation 0.1 let $(\mathcal{F}_n)$ be an object of $Coh(U, I\mathcal{O}_U)$. Let M be as in (equation guess (Tag 0EHD)). Assume*

(a) *the inverse system $H^0(U, \mathcal{F}_n)$ has Mittag-Leffler,*
(b) *the limit topology on M agrees with the I-adic topology, and*
(c) *the image of $M \to H^0(U, \mathcal{F}_n)$ is a finite A-module for all n.*

Then $(\mathcal{F}_n)$ extends canonically to X. In particular, if A is I-adically complete, then $(\mathcal{F}_n)$ is the completion of a coherent $\mathcal{O}_U$-module.

证明. Since $H^0(U, \mathcal{F}_n)$ has the Mittag-Leffler condition and since the limit topology on M is the I-adic topology we see that $\{M/I^nM\}$ and $\{H^0(U, \mathcal{F}_n)\}$ are pro-isomorphic inverse systems of A-modules. Thus if we set

$$\mathcal{G}_n = \widetilde{M/I^nM}$$

then we see that to verify the condition in Definition 0.5 it suffices to show that M is a finite module over the I-adic completion of A. This follows from the fact that M/I^nM is finite by condition (c) and the above and Algebra, Lemma lemma finite over complete ring (Tag 031D). □

Definition 0.5. In Situation 0.1 let $(\mathcal{F}_n)$ be an object of $Coh(U, I\mathcal{O}_U)$. We say $(\mathcal{F}_n)$ *canonically extends to* X if the the inverse system

$$\{\widetilde{H^0(U, \mathcal{F}_n)}\}_{n\geq 1}$$

in $QCoh(\mathcal{O}_X)$ is pro-isomorphic to an object $(\mathcal{G}_n)$ of $Coh(X, I\mathcal{O}_X)$.

Primary problem: Tag 0EJ9

Lemma 0.6. *In Situation 0.1 let $(\mathcal{F}_n)$ be an object of $Coh(U, I\mathcal{O}_U)$. Assume*

1. *A is local and $\mathfrak{a} = \mathfrak{m}$ is the maximal ideal,*
2. *A has a dualizing complex,*
3. *$I = (f)$ is a principal ideal,*
4. *$(\mathcal{F}_n)$ satisfies the $(2, 3)$-inequalities.*

Then $(\mathcal{F}_n)$ extends to X. In particular, if A is I-adically complete, then $(\mathcal{F}_n)$ is the completion of a coherent $\mathcal{O}_U$-module.

证明. Recall that $Coh(U, I\mathcal{O}_U)$ is an abelian category, see Cohomology of Schemes, Lemma lemma inverse systems abelian (Tag 087X). Over affine opens of U the object $(\mathcal{F}_n)$ corresponds to a finite module over a Noetherian ring (Cohomology of Schemes, Lemma lemma inverse systems affine (Tag 087W)). Thus the kernels of the maps $f^N : (\mathcal{F}_n) \to (\mathcal{F}_n)$ stabilize for N large enough. By Lemmas lemma map kernel cokernel on closed (Tag 0EIU) and lemma essential image completion (Tag 0EIL) in order to prove the lemma we may replace $(\mathcal{F}_n)$ by the image of such a map. Thus we may assume f is injective on $(\mathcal{F}_n)$. After this replacement the equivalent conditions of Lemma lemma equivalent f good (Tag 0EH9) hold for the inverse system $(\mathcal{F}_n)$ on U. We will use this without further mention in the rest of the proof.

We will check hypotheses (a), (b), and (c) of Lemma 0.4. Hypothesis (b) holds by Cohomology, Lemma lemma topology I adic f (Tag 0EHA).

Pick an inverse system of modules $\{M_n\}$ as in Lemma lemma system of modules (Tag 0DXT). We may assume $H^0_{\mathfrak{m}}(M_n) = 0$ by replacing M_n by $M_n/H^0_{\mathfrak{m}}(M_n)$ if necessary. Then we obtain short exact sequences

$$0 \to M_n \to H^0(U, \mathcal{F}_n) \to H^1_{\mathfrak{m}}(M_n) \to 0$$

for all n. Let E be an injective hull of the residue field of A. By Lemma 0.3 and our current assumption (4) we can choose, an integer $m \geq 0$, finite A-modules N_1 and N_2 annihilated by f^c for some $c \geq 0$ and compatible systems of maps

$$H^i_{\mathfrak{m}}(M_n) \to \mathrm{Hom}_A(N_i, E), \quad i = 1, 2$$

for $n \geq m$ with the properties stated in the lemma.

We know that $M = \lim H^0(U, \mathcal{F}_n)$ is an A-module whose limit topology is the f-adic topology. Thus, given n, the module M/f^nM is a subquotient of $H^0(U, \mathcal{F}_N)$ for some $N \gg n$. Looking at the information obtained above we see that f^cM/f^nM is a finite A-module. Since f is a nonzerodivisor on M we conclude that $M/f^{n-c}M$ is a finite A-module. In this way we see that hypothesis (c) of Lemma 0.4 holds.

Next, we study the module

$$Ob = \lim H^1(U, \mathcal{F}_n) = \lim H^2_{\mathfrak{m}}(M_n)$$

For $n \geq m$ let K_n be the kernel of the map $H^2_{\mathfrak{m}}(M_n) \to \mathrm{Hom}_A(N_2, E)$. Set $K = \lim K_n$. We obtain an exact sequence

$$0 \to K \to Ob \to \mathrm{Hom}_A(N_2, E)$$

By the above the limit topology on $Ob = \lim H^2_{\mathfrak{m}}(M_n)$ is the f-adic topology. Since N_2 is annihilated by f^c we conclude the same is true for the limit topology on $K = \lim K_n$. Thus K/fK is a subquotient of K_n for $n \gg 1$. However, since $\{K_n\}$ is pro-isomorphic to an inverse system of finite length A-modules (by the conclusion of Lemma 0.3) we conclude that K/fK is a subquotient of a finite length A-module. It follows that K is a finite A-module, see Algebra, Lemma lemma finite over complete ring (Tag 031D). (In fact, we even see that $\dim(\mathrm{Supp}(K)) = 1$ but we will not need this.)

Given $n \geq 1$ consider the boundary map

$$\delta_n : H^0(U, \mathcal{F}_n) \longrightarrow \lim_N H^1(U, f^n\mathcal{F}_N) \xrightarrow{f^{-n}} Ob$$

(the second map is an isomorphism) coming from the short exact sequences

$$0 \to f^n\mathcal{F}_N \to \mathcal{F}_N \to \mathcal{F}_n \to 0$$

For each n set

$$P_n = \mathrm{Im}(H^0(U, \mathcal{F}_{n+m}) \to H^0(U, \mathcal{F}_n))$$

where m is as above. Observe that $\{P_n\}$ is an inverse system and that the map $f : \mathcal{F}_n \to \mathcal{F}_{n+1}$ on global sections maps P_n into P_{n+1}. If $p \in P_n$, then $\delta_n(p) \in K \subset Ob$ because $\delta_n(p)$ maps to zero in $H^1(U, f^n\mathcal{F}_{n+m}) = H^2_{\mathfrak{m}}(M_m)$ and the composition of δ_n and $Ob \to \mathrm{Hom}_A(N_2, E)$ factors through $H^2_{\mathfrak{m}}(M_m)$ by our choice of m. Hence

$$\bigoplus\nolimits_{n \geq 0} \mathrm{Im}(P_n \to Ob)$$

is a finite graded $A[T]$-module where T acts via multiplication by f. Namely, it is a graded submodule of $K[T]$ and K is finite over A. Arguing as in the proof of Cohomology, Lemma lemma ML general (Tag 0GYK)[1] we find that the inverse system $\{P_n\}$ satisfies ML. Since $\{P_n\}$ is pro-isomorphic to $\{H^0(U, \mathcal{F}_n)\}$ we conclude that $\{H^0(U, \mathcal{F}_n)\}$ has ML. Thus hypothesis (a) of Lemma 0.4 holds and the proof is complete. □

[1] Choose homogeneous generators of the form $\delta_{n_j}(p_j)$ for the displayed module. Then if $k = \max(n_j)$ we find that for $n \geq k$ and any $p \in P_n$ we can find $a_j \in A$ such that $p - \sum a_j f^{n-n_j} p_j$ is in the kernel of δ_n and hence in the image of $P_{n'}$ for all $n' \geq n$. Thus $\mathrm{Im}(P_n \to P_{n-k}) = \mathrm{Im}(P_{n'} \to P_{n-k})$ for all $n' \geq n$.